

CS 771A: Intro to Machine Learning, IIT Kanpur			Endsem Exam (30 Apr 2026)	
Name	MELBO			40 marks Page 1 of 4
Roll No	260007	Dept.	AWSM	

Instructions:

1. This question paper contains 2 pages (4 sides of paper). Please verify.
2. Write your name, roll number, department in **block letters** with **ink** on **each page**.
3. Write your final answers neatly **with a blue/black pen**. Pencil marks may get smudged.
4. Don't overwrite/scratch answers especially in MCQ – ambiguous cases will get 0 marks.



Q1 (Kernel Kontortions) $K_i, i = 1, 2$ are Mercer kernels $K_i: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ with maps $\phi_i: \mathbb{R}^2 \rightarrow \mathbb{R}^D$ for $D \gg 2$ i.e. for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2, K_i(\mathbf{x}, \mathbf{y}) = \langle \phi_i(\mathbf{x}), \phi_i(\mathbf{y}) \rangle$. Design maps for 4 new kernels created using K_1, K_2 . You may use the notation $[\mathbf{u}, \mathbf{v}, \mathbf{w}]$ to denote vector concatenation. Your maps must be in terms of the maps $\phi_i(\mathbf{x}), i = 1, 2$, and universal constants, and not contain any $\mathbf{y}$ term e.g. $\frac{1}{\phi_1(\mathbf{x})^\top \phi_1(\mathbf{x})} \cdot [6 \cdot \phi_1(\mathbf{x}), \sqrt{7} \cdot \phi_2(\mathbf{x}), 3] \in \mathbb{R}^{2D+1}$ is a valid map. No derivation. **(4 x 2 = 8 marks)**

Part 1. $K_3(\mathbf{x}, \mathbf{y}) = K_1(\mathbf{x}, \mathbf{y}) + 3$

$$\phi_3(\mathbf{x}) = [\phi_1(\mathbf{x}), \sqrt{3}]$$

Part 2. $K_4(\mathbf{x}, \mathbf{y}) = \frac{4 \cdot K_2(\mathbf{x}, \mathbf{y})}{K_2(\mathbf{x}, \mathbf{x}) \cdot K_2(\mathbf{y}, \mathbf{y})}$

$$\phi_4(\mathbf{x}) = \frac{2}{\phi_2(\mathbf{x})^\top \phi_2(\mathbf{x})} \cdot \phi_2(\mathbf{x})$$

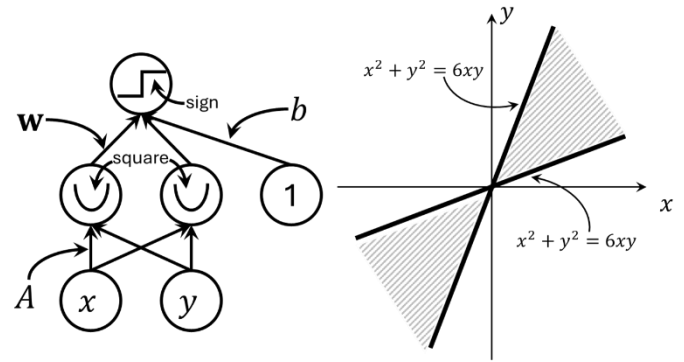
Part 3. $K_5(\mathbf{x}, \mathbf{y}) = 5 \cdot K_1(\mathbf{x}, \mathbf{y}) + 10 \cdot K_2(\mathbf{x}, \mathbf{y})$

$$\phi_5(\mathbf{x}) = [\sqrt{5} \cdot \phi_1(\mathbf{x}), \sqrt{10} \cdot \phi_2(\mathbf{x})]$$

Part 4. $K_6(\mathbf{x}, \mathbf{y}) = 6 \cdot K_1(\mathbf{x}, \mathbf{x}) \cdot K_2(\mathbf{x}, \mathbf{y}) \cdot K_1(\mathbf{y}, \mathbf{y})$

$$\phi_6(\mathbf{x}) = \sqrt{6} \cdot \phi_1(\mathbf{x})^\top \phi_1(\mathbf{x}) \cdot \phi_2(\mathbf{x})$$

Q2 (Scissor Networks) Melba wants to use a neural network with architecture shown on the left, to solve a binary classification problem shown on the right. The bold decision boundaries in the figure depict the pair-of-lines conic $x^2 + y^2 = 6xy$. The NN parameters are a 2×2 matrix $A \in \mathbb{R}^{2 \times 2}$, a 2D vector $\mathbf{w} \in \mathbb{R}^2$ and a bias $b \in \mathbb{R}$. The output of the NN is $\text{sign}(\mathbf{w}^\top \phi(\mathbf{x}) + b)$ where $\phi(\mathbf{x}) = (A\mathbf{x})^2$



with square activation being applied coordinate-wise i.e. for a vector $\mathbf{v} = (v_1, v_2)$, the square activation is applied as $(\mathbf{v})^2 \stackrel{\text{def}}{=} (v_1^2, v_2^2)$. Find values of the parameters $A, \mathbf{w}, b$ so that the NN gives output -1 in the shaded region and $+1$ in the white region. Be careful to write value of A and not $A^\top$ (note that we have $\phi(\mathbf{x}) = (A\mathbf{x})^2$ not $(A^\top \mathbf{x})^2$). **Write only in the space provided. (7 marks)**

$$A = \begin{bmatrix} \boxed{1} & \boxed{1} \\ \boxed{1} & \boxed{-1} \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} \boxed{-1} & \boxed{2} \end{bmatrix} \quad b = \boxed{0}$$

Q3. (Matrix Norm?) $A \in \mathbb{R}^{m \times m}$ is a (possibly non-symmetric) square matrix with SVD $A = U\Sigma V^\top$. $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_m) \in \mathbb{R}^{m \times m}$ with $\sigma_1 > \sigma_2 > \dots > \sigma_m > 0$ (all singular values are non-zero and distinct), $U = [\mathbf{u}_1, \dots, \mathbf{u}_m], V = [\mathbf{v}_1, \dots, \mathbf{v}_m] \in \mathbb{R}^{m \times m}$ are orthonormal. Let $b^* \stackrel{\text{def}}{=} \min_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2$. Note the constraint $\|\mathbf{x}\|_2 = 1$. Find the value of b^* in terms of $\mathbf{u}_i, \mathbf{v}_i, \sigma_i, i \in [m]$. Show brief derivation. *Hint: try expressing $\mathbf{x}$ as a linear combination of the right singular vectors $\mathbf{v}_1, \dots, \mathbf{v}_m$ which is always possible as they form a basis over $\mathbb{R}^m$. You may also find $VV^\top = I = V^\top V$ useful. (2+3 = 5 marks)*

$$b^* \stackrel{\text{def}}{=} \min_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2 = \sigma_m \text{ or equivalently, } \min_i \{\sigma_i\}$$

Give brief derivation below

Let $\mathbf{z} \stackrel{\text{def}}{=} V^\top \mathbf{x} \in \mathbb{R}^m$ be representation of $\mathbf{x}$ in the basis of the right singular vectors i.e. $\mathbf{x} = V\mathbf{z}$. Note that $\|\mathbf{z}\|_2^2 = \mathbf{z}^\top \mathbf{z} = \mathbf{x}^\top V V^\top \mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|_2^2 = 1$. Thus, $\|A\mathbf{x}\|_2 = \|U\Sigma V^\top V\mathbf{z}\|_2 = \|U\Sigma\mathbf{z}\|_2$. This is unsurprising since all that V can do is rotate and swap stuff and not change the length of a vector. By using orthonormality of U similarly, we get $\|U\Sigma\mathbf{z}\|_2 = \|\Sigma\mathbf{z}\|_2$.

Thus, the problem reduces to $\min_{\|\mathbf{z}\|_2=1} \|\Sigma\mathbf{z}\|_2 = \min_{\|\mathbf{z}\|_2=1} \sqrt{\sum_{i \in [m]} \sigma_i^2 z_i^2}$.

Since $\sigma_1 > \sigma_2 > \dots > \sigma_m$ and $z_i^2 \geq 0$ for all i , we get $\sum_{i \in [m]} \sigma_i^2 z_i^2 \geq \sigma_m^2 (\sum_{i \in [m]} z_i^2) = \sigma_m^2$. This maximum value is achieved when $\mathbf{z} = (0, 0, \dots, 1) = \mathbf{e}_m$ i.e. when $\mathbf{x} = \mathbf{v}_m$, the leading right singular vector. Indeed, we have $\|A\mathbf{x}\|_2 = \|U\Sigma V^\top \mathbf{v}_m\|_2 = \|U\Sigma \mathbf{e}_m\|_2 = \sigma_m \cdot \|\mathbf{e}_m\|_2 = \sigma_m$.

This quantity is often called the *inf-norm* although it is actually not a norm – can you tell which property of norms does it violate?

CS 771A: Intro to Machine Learning, IIT Kanpur			Endsem Exam (30 Apr 2026)	
Name	MELBO			40 marks Page 3 of 4
Roll No	260007	Dept.	AWSM	

Q4 (Mending Melbo's Mistake – Part III) In the midsem, we derived the dual problem to maximize the mean margin with data $\mathbf{x}^i \in \mathbb{R}^d, y^i \in \{-1, +1\}$. We will now solve it using coordinate descent.

$$\min \frac{1}{4\beta} \cdot \left\| \sum_{i \in [N]} (1 + \alpha_i) \cdot y^i \cdot \mathbf{x}^i \right\|_2^2 + \beta$$

s.t. $\beta \geq 0$ and $\alpha_i \geq 0 \quad \forall i \in [n]$

(Part 1) Give optimal value of $\beta \geq 0$ after fixing all $\alpha_i, i \in [n]$. No derivation needed. (4 marks)

$$\beta = \frac{1}{2} \cdot \left\| \sum_{i \in [N]} (1 + \alpha_i) \cdot y^i \cdot \mathbf{x}^i \right\|_2$$

Setting the (partial) derivative of the objective (w.r.t. β) to zero itself a solution that naturally satisfies the positivity constraint. Thus, it must be the optimal constrained solution too.

(Part 2) Give optimal value of $\alpha_1 \geq 0$ after fixing β and all $\alpha_i, i \geq 2$. No derivation. (6 marks)

$$\alpha_1 = \max \left\{ 0, -1 - \frac{\sum_{i \geq 2} (1 + \alpha_i) \cdot y^1 y^i \cdot (\mathbf{x}^1)^\top \mathbf{x}^i}{\|\mathbf{x}^1\|_2^2} \right\}$$

Let $\mathbf{p} \stackrel{\text{def}}{=} y^1 \cdot \mathbf{x}^1$ and $\mathbf{q} \stackrel{\text{def}}{=} \mathbf{p} + \sum_{i \geq 2} (1 + \alpha_i) \cdot y^i \cdot \mathbf{x}^i$. The problem, restricted to α_1 looks like

$$\min_{\alpha_1 \geq 0} \|\alpha_1 \cdot \mathbf{p} + \mathbf{q}\|_2^2 = \min_{\alpha_1 \geq 0} \alpha_1^2 \cdot \|\mathbf{p}\|_2^2 + 2\alpha_1 \cdot \mathbf{p}^\top \mathbf{q}. \text{ The QUIN trick gives } \alpha_1 = \max \left\{ 0, -\frac{\mathbf{p}^\top \mathbf{q}}{\|\mathbf{p}\|_2^2} \right\}$$

Q5 (Regretrons) The seminal result of Minsky-Papert showed that perceptrons cannot solve some simple binary classification problems e.g. XOR. Melbo wants to show a similar result for regression problems. Given data $(x, y) \in \mathbb{R} \times \mathbb{R}$ with $x \in [-1, 1], y = x^2$, Melbo wants to show that no linear model can achieve zero least squares error (even on train data). Help Melbo do this in two steps.

(Step 1) Find the least squares error of the linear model $ax + b$ over the interval $[-1, 1]$ by solving the definite integral $\mathcal{L}(a, b) \stackrel{\text{def}}{=} \frac{1}{2} \int_{-1}^1 (ax + b - x^2)^2 dx$. To give your answer, write the coefficients of various terms such as ab, a^2 , etc in the blanks (the last blank is the constant term). Your answers must be universal constants e.g. $\frac{1}{4}, 12, 0$, etc and must not contain x, y, a, b . If a certain term does not appear in the solution, write its coefficient as zero (blank cells receive no marks). (6 marks)

$$\mathcal{L}(a, b) = \boxed{1/3} \cdot a^2 + \boxed{1} \cdot b^2 +$$

$$\boxed{0} \cdot ab + \boxed{0} \cdot a + \boxed{-2/3} \cdot b + \boxed{1/5}$$

(Step 2) Find the values of $a, b \in \mathbb{R}$ that minimize the loss $\mathcal{L}(a, b)$. Write down the value of a, b that achieve the lowest loss and the lowest loss value. No derivation needed. (1+1+2=4 marks)

$$a = \boxed{0}, b = \boxed{1/3}$$

$$\min_{a, b \in \mathbb{R}} \mathcal{L}(a, b) = \boxed{4/45}$$

Note: the space below is for rough work only. Content written below will not be evaluated.

ROUGH WORK AREA
NOT TO BE EVALUATED

ROUGH WORK AREA
NOT TO BE EVALUATED

ROUGH WORK AREA
NOT TO BE EVALUATED